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(57) Abstract:

Quantum–Classical Chiplet Architecture for Scalable Hybrid Computing Systems. The present invention relates to a hybrid quantum–classical computing architecture implemented using chiplet-based semiconductor integration. The architecture comprises classical processor chiplets, a cryogenic control chiplet, and a quantum processing unit chiplet interconnected through a silicon interposer substrate within a system-in-package configuration. The interposer architecture incorporates high-density interconnect routing and thermal isolation structures to maintain separation between ambient temperature classical processing domains and cryogenic quantum processing domains. The system further includes a spectral veto control module configured to analyze spectral characteristics of classical control signals and selectively enable or prevent execution of quantum instructions when signal noise exceeds predefined thresholds. The architecture enables efficient hybrid quantum–classical computation by reducing communication latency and maintaining signal integrity between classical and quantum processing components. [FIG. 1]

FORM 2

THE PATENT ACT 1970

(39 OF 1970)

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THE PATENT RULES, 2003

COMPLETE SPECIFICATION

(SEE SECTION 10)

1. TITLE OF THE INVENTION

QUANTUM–CLASSICAL CHIPLET ARCHITECTURE FOR SCALABLE HYBRID COMPUTING SYSTEMS

2. APPLICANT (S)

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3. PREAMBLE TO THE DESCRIPTION

THE FOLLOWING SPECIFICATION PARTICULARLY DESCRIBES THE INVENTION AND THE MANNER IN WHICH IT IS TO BE PERFORMED.

4. DESCRIPTION

QUANTUM–CLASSICAL CHIPLET ARCHITECTURE FOR SCALABLE HYBRID COMPUTING SYSTEMS.

TECHNICAL FIELD

[0001] The present invention relates generally to heterogeneous computing architectures and semiconductor system integration. More particularly, the invention relates to a hybrid quantum–classical computing architecture implemented using chiplet-based integration within a system-in-package configuration. The invention further relates to semiconductor systems in which classical processor chiplets and quantum processing unit chiplets are interconnected through a silicon interposer substrate incorporating thermal isolation structures and signal integrity control mechanisms to enable low-latency hybrid quantum–classical computation.

BACKGROUND OF THE INVENTION

[0002] Quantum computing has emerged as a promising computational paradigm capable of solving certain classes of problems that are difficult or computationally expensive for conventional classical computers. Quantum processors operate using quantum bits, or qubits, which exploit phenomena such as superposition and entanglement to perform specialized computational tasks.

[0003] In practical implementations, quantum processors typically operate in conjunction with classical computing systems. Classical processors perform tasks such as program execution, control signal generation, algorithm orchestration, and processing of measurement results obtained from quantum operations. Consequently, hybrid quantum–classical computing architectures are required for the effective deployment of quantum computing technologies.

[0004] Existing hybrid computing systems generally employ loosely coupled architectures in which classical processors communicate with quantum processors through external interfaces such as peripheral buses, network links, or dedicated control hardware. These configurations often introduce communication latency and limit the efficiency of real-time hybrid quantum–classical algorithms.

[0005] Another challenge associated with hybrid quantum–classical integration arises from the differing operating environments of classical and quantum processors. Classical processors typically operate at ambient temperatures and generate thermal energy and electromagnetic noise. Quantum processors, however, often require cryogenic environments in order to maintain qubit coherence and stability. Integrating classical and quantum processing components therefore requires careful management of thermal isolation and signal integrity.

[0006] Conventional approaches typically maintain physical separation between classical control hardware and quantum processing systems in order to preserve

cryogenic conditions. While this separation reduces thermal interference, it increases communication latency and reduces the efficiency of tightly coupled hybrid computational workflows.

[0007] Accordingly, there exists a need for improved semiconductor architectures capable of integrating classical processors and quantum processors within a unified system while maintaining thermal isolation, signal integrity, and efficient communication between computing domains.

SUMMARY OF THE INVENTION

[0008] An object of the present invention is to provide a hybrid quantum–classical computing architecture capable of integrating classical processors and quantum processors within a unified semiconductor package.

[0009] Another object of the invention is to provide a chiplet-based system architecture in which classical processor chiplets and quantum processing unit chiplets are interconnected through a silicon interposer substrate within a system-in-package configuration.

[0010] A further object of the invention is to provide a semiconductor interposer architecture that enables high-density electrical interconnection and low-latency communication between classical computing components and quantum processing components.

[0011] Another object of the invention is to provide thermal isolation mechanisms within the interposer structure to maintain separation between ambient temperature classical processing domains and cryogenic quantum processing domains.

[0012] A further object of the invention is to provide a spectral veto control mechanism configured to monitor spectral characteristics of classical control signals and selectively enable or prevent execution of quantum instructions when noise levels exceed predefined thresholds.

[0013] Accordingly, the invention provides a heterogeneous hybrid computing architecture comprising classical processor chiplets, quantum processing unit chiplets, cryogenic control chiplets, and associated subsystems integrated within a silicon interposer substrate incorporating thermal isolation structures and signal integrity control mechanisms, thereby enabling efficient and scalable hybrid quantum–classical computation.

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 illustrates a system architecture of a hybrid quantum–classical chiplet computing system including classical processor chiplets, a cryogenic control chiplet, a silicon interposer fabric, a quantum processing unit chiplet, a quantum error correction chiplet, and a spectral veto control module configured to enable coordinated hybrid quantum-classical computation.

[0015] FIG. 2 illustrates a silicon interposer chiplet integration layout showing placement of classical CPU and GPU chiplets, a cryogenic control chiplet, a quantum processing unit chiplet, and a quantum error correction chiplet interconnected through a high-density interconnect fabric and through-silicon vias within a system-in-package configuration.

[0016] FIG. 3 illustrates a thermal isolation architecture for the hybrid quantum–classical chiplet system showing separation between an ambient temperature domain and a cryogenic domain using thermal isolation trenches, micro-vacuum gaps, and superconducting through-silicon interconnect structures.

[0017] FIG. 4 illustrates a memory-mapped hybrid quantum execution pipeline in which classical program execution triggers quantum instruction detection, parameter transfer through shared interposer memory, cryogenic control signal translation, quantum gate execution, measurement acquisition, quantum error correction processing, and result retrieval by classical processors.

[0018] FIG. 5 illustrates a dual-band spectral veto control mechanism configured to analyze spectral characteristics of classical control signals, determine signal-to-noise ratios, compare the ratios with predefined thresholds, and selectively enable or veto execution of quantum instructions before transmission to the cryogenic control module.

DETAILED DESCRIPTION

[0019] The invention provides a heterogeneous hybrid computing architecture integrating classical processor chiplets and quantum processing chiplets within a system-in-package configuration. Referring to FIG. 1, the system includes a classical CPU chiplet (100) and a classical GPU chiplet (102) configured to execute classical program instructions and manage hybrid quantum-classical computational workflows. These classical processors generate control instructions and computational tasks that may include quantum operations requiring execution within a quantum processing environment.

[0020] A cryogenic control chiplet (104) operates as an interface between the classical computing domain and a quantum processing domain. The cryogenic control chiplet receives instruction parameters from the classical processor chiplets and translates these parameters into control signals suitable for operating quantum circuits within a quantum processing unit. The cryogenic control chiplet further manages timing synchronization and signal conditioning required for quantum gate execution.

[0021] The system incorporates a silicon interposer fabric (106) configured to interconnect multiple chiplets within a high-density system-in-package architecture. The interposer provides high-bandwidth electrical communication pathways between the classical processor chiplets, the cryogenic control chiplet, and quantum processing components. As illustrated in FIG. 2, the silicon interposer substrate (200) supports integration of classical CPU chiplets (202), classical GPU chiplets (204), a quantum

processing unit chiplet (208), and a quantum error correction chiplet (210). High-density interconnect routing structures (212) and through-silicon vias (214) enable efficient communication between the integrated chiplets.

[0022] The quantum processing unit chiplet (108) performs quantum computational operations using qubits and quantum gate sequences generated by the cryogenic control chiplet. Measurement results generated by the quantum processing unit are transferred to a quantum error correction chiplet (110) which performs error correction processing to mitigate noise and maintain reliability of quantum computations.

[0023] Referring to FIG. 3, the system includes a thermal isolation architecture designed to maintain separation between classical processing domains and quantum processing domains. Classical processor chiplets operate within an ambient temperature domain (300), while the quantum processing unit chiplet operates within a cryogenic domain (312). A silicon interposer layer (306) incorporates deep trench thermal isolation structures (308) and micro-vacuum thermal gaps (310) that reduce thermal conduction between the ambient temperature region and the cryogenic region. Superconducting through-silicon interconnect structures (316) provide electrical connectivity while minimizing heat transfer and electromagnetic interference.

[0024] Referring to FIG. 4, the invention further provides a memory-mapped hybrid execution pipeline enabling coordinated hybrid quantum-classical computation. A classical program execution stage (400) generates instructions executed by the classical

processor chiplets. A quantum instruction detection stage (402) identifies instructions that require quantum computation. Parameters associated with the detected quantum instructions are transferred to shared interposer memory through a direct memory access transfer stage (404). The cryogenic control chiplet converts these parameters into cryogenic control signals (406) that initiate quantum gate execution within the quantum processing unit (408). Measurement results are obtained in a measurement acquisition stage (410) and processed through a quantum error correction stage (412). The processed results are written back to shared memory (414) and retrieved by classical processors through a processor readout stage (416).

[0025] Referring to FIG. 5, the system further includes a spectral veto control mechanism designed to prevent noisy control signals from affecting quantum computation. Classical control signals are received at a classical control signal input stage (500). A spectral analysis engine (502) evaluates the spectral characteristics of the control signals, and a noise energy measurement module (504) determines the signal-to-noise ratio associated with the signal. A noise threshold comparator (506) compares the signal-to-noise ratio with a predefined threshold. When the signal-to-noise ratio exceeds the threshold, a quantum instruction enable module (508) allows the instruction to proceed to the cryogenic control module (512). When the signal-to-noise ratio falls below the threshold, a quantum instruction veto module (510) prevents execution of the quantum instruction to protect the quantum processing unit from noise-induced errors.

[0026] Through the above described architecture, the invention provides a scalable hybrid quantum-classical computing platform capable of integrating classical and quantum processors within a unified semiconductor package while maintaining thermal isolation, signal integrity, and low-latency communication between computational domains.

5. CLAIMS

I/We Claim:

1. A heterogeneous hybrid computing system comprising:

a plurality of classical processor chiplets configured to execute classical computing instructions;

at least one quantum processing unit (QPU) chiplet configured to perform quantum computational operations on qubits;

a silicon interposer substrate configured to electrically and physically interconnect the plurality of classical processor chiplets and the quantum processing unit chiplet within a system-in-package architecture;

a cryogenic control chiplet configured to translate classical instruction data into quantum control signals for operating the quantum processing unit chiplet;

one or more thermal isolation structures formed within the silicon interposer substrate to maintain thermal separation between the classical processor chiplets operating at ambient temperature and the quantum processing unit chiplet operating at cryogenic temperature;

a memory-mapped communication interface configured to enable direct data exchange between the classical processor chiplets and the quantum processing unit chiplet through the silicon interposer substrate; and

a spectral veto control module configured to monitor spectral characteristics of control signals generated by the classical processor chiplets and selectively prevent execution of quantum operations when noise levels exceed a predefined threshold,

wherein the silicon interposer substrate enables ultra-low-latency communication between the classical processor chiplets and the quantum processing unit chiplet to facilitate hybrid quantum-classical computation.

2. The system as claimed in claim 1, wherein the silicon interposer substrate comprises a high-density interconnect fabric configured to provide electrical communication between the classical processor chiplets, the cryogenic control chiplet, and the quantum processing unit chiplet.
3. The system as claimed in claim 1, wherein the silicon interposer substrate further comprises through-silicon vias configured to provide vertical electrical interconnections between multiple chiplets integrated within the system-in-package architecture.
4. The system as claimed in claim 1, wherein the thermal isolation structures comprise deep trench isolation regions formed within the silicon interposer substrate.
5. The system as claimed in claim 1, wherein the thermal isolation structures further comprise micro-vacuum gaps configured to reduce thermal conduction between the ambient temperature domain and the cryogenic domain.
6. The system as claimed in claim 1, wherein the silicon interposer substrate further comprises superconducting through-silicon interconnects configured to reduce electromagnetic interference and thermal conduction between chiplets.
7. The system as claimed in claim 1, further comprising a quantum error correction chiplet configured to receive measurement results from the quantum processing unit chiplet and perform real-time quantum error correction processing.
8. The system as claimed in claim 1, wherein the spectral veto control module comprises a spectral analysis engine configured to analyze frequency components of classical control signals generated by the classical processor chiplets.
9. The system as claimed in claim 8, wherein the spectral veto control module further comprises a noise measurement module configured to determine a signal-to-noise ratio of the analyzed control signals.
10. The system as claimed in claim 9, wherein the spectral veto control module further comprises a threshold comparator configured to enable quantum instruction execution when the signal-to-noise ratio exceeds a predefined threshold and to veto the quantum instruction when the signal-to-noise ratio falls below the predefined threshold.

6. SIGNATURE WITH DATE

SWAPNA SUNDAR
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7. ABSTRACT

Quantum–Classical Chiplet Architecture for Scalable Hybrid Computing Systems. The present invention relates to a hybrid quantum–classical computing architecture implemented using chiplet-based semiconductor integration. The architecture comprises classical processor chiplets, a cryogenic control chiplet, and a quantum processing unit chiplet interconnected through a silicon interposer substrate within a system-in-package configuration. The interposer architecture incorporates high-density interconnect routing and thermal isolation structures to maintain separation between ambient temperature classical processing domains and cryogenic quantum processing domains. The system further includes a spectral veto control module configured to analyze spectral characteristics of classical control signals and selectively enable or prevent execution of quantum instructions when signal noise exceeds predefined thresholds. The architecture enables efficient hybrid quantum–classical computation by reducing communication latency and maintaining signal integrity between classical and quantum processing components. [FIG. 1]

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